Deepanjay Nandal

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Summary

Senior Software Engineer with 3+ years building systems, from CVE validation for Google's Android Security team to LLM inference caching and event-driven pipelines processing 100k+ daily events. Combines low-level systems and security depth with AI infrastructure across Kubernetes, AI agents, semantic caching, hybrid retrieval, and Kafka/Redis.

Experience

Discover Excellence LLC — California, USA
Senior Software Engineer (Contract) — *Jan 2026 – Jun 2026*

- Built a high-throughput Kafka-based visitor intelligence pipeline processing 100k+ daily behavioral events at sub-40ms p95 latency, identifying high-intent customer activity across multi-tenant eCommerce stores.
- Designed a Python-based AI sales agent using GPT-4.1 mini and tool calling to evaluate customer behavior, select personalized offers, and orchestrate asynchronous outreach via Celery, Redis, and AWS SES.
- Served personalized product feeds at the edge via Cloudflare, backed by Redis collaborative filtering, achieving 92% cache hit rate and sub-30ms p95 latency across multi-tenant deployments.

Ittiam Systems Pvt Ltd — Bengaluru, India
Senior Software Engineer — *Mar 2022 – Jun 2024*

- Authored 50+ CTS/STS validation tests across multiple Android versions, merged into the Android Open Source Project (AOSP), each proving a CVE patch, working directly with Google engineers.
- Reverse-engineered stripped Bluetooth binaries in Ghidra to locate internal functions with no exposed symbol, authoring CTS/STS tests for Bluetooth CVEs Google had marked infeasible.
- Hooked inaccessible Android methods at runtime with Frida to reach Java-layer vulnerable code tests could not call directly, reproducing and validating CVEs the team could not otherwise reach.
- Analyzed vendor firmware at the assembly level in Ghidra and IDA Pro to verify security-patch presence across OEM builds, comparing disassembly against upstream AOSP fixes.
- Triaged ClusterFuzz crash reports by reproducing crashes and tracing execution to the exact fault location, documenting root-cause analysis for reviewer verification.

Projects

BitMod (AI Inference Cache) | *Python, FastAPI, Vector Embeddings, Multi-tenant, Docker* — *Jan – Jun 2026*

- Architected a drop-in 9-layer LLM cache proxy with SHA-256 exact lookup, semantic reuse, and Bayesian evidence scoring, achieving 76% hit rate on a 50-query benchmark (100% on repeats, 53% on rephrasings).
- Held zero false serves across 375 held-out queries with nothing cached, reached 90–101x speedup over self-hosted LLM calls at sub-90ms cached, and enforced multi-tenant namespace isolation in SQL.
- Designed cost-aware LRU eviction scoring entries by predicted future hits against original LLM spend.

CaseTally (Legal Search Platform) | *FastAPI, pgvector, RAG, Kubernetes, KEDA* — *Jun – Dec 2025*

- Constructed a RAG over 83k+ U.S. Code chunks with LLM issue decomposition, fusing PostgreSQL full-text and pgvector HNSW via RRF at MRR 0.85 and P@3 0.78 while cutting p50 latency from 325ms to 45ms (7x).
- Migrated the stack to Kubernetes with a Postgres StatefulSet, Traefik ingress, probes and advisory-locked ingestion Job, achieving zero-downtime rollouts (240 requests, 0 failures) and a 73% smaller backend image.
- Built an embedding queue on FOR UPDATE SKIP LOCKED with KEDA scaling workers 0 to 3 on queue depth, verifying 3,000/3,000 rows with zero loss or duplication under SIGKILL and raising worker throughput 55%.

Elastic Face Recognition (Cloud ML Pipeline) | *AWS EC2, Lambda, SQS, PyTorch* — *Aug – Dec 2025*

- Scaled MTCNN/FaceNet inference across 15 EC2 instances with a custom SQS queue-depth autoscaler counting in-flight messages to prevent premature scale-in, sustaining 0.96s latency under 100 concurrent requests.
- Re-architected into an event-driven serverless pipeline with Lambda workers on SQS triggers and Docker images on ECR, adding an IoT Greengrass edge tier that short-circuits no-face cases, cutting response 2.3x to 776ms.

Technical Skills

Languages: Python, Java, SQL
Backend Technologies: Distributed Systems, Microservices, Event-Driven Architecture, Multi-tenancy, Concurrency Control, REST APIs, WebSockets, Message Queues, Rate Limiting
Data & Storage: Apache Kafka, PostgreSQL, SQLite, Redis, pgvector, HNSW, Distributed Caching
Security & Systems: Android Open Source Project (AOSP), Binary Analysis, Ghidra, Frida, IDA Pro
Cloud & DevOps: Kubernetes, KEDA, Docker, AWS (EC2, Lambda, SQS, IoT Greengrass), Cloudflare
AI Infrastructure: Retrieval-Augmented Generation (RAG), Vector Search, Reciprocal Rank Fusion (RRF), Large Language Model (LLM) APIs, Semantic Caching, Embeddings, Collaborative Filtering, Retrieval Evaluation

Education

Arizona State University — Tempe, Arizona, USA
Master of Science (M.S.) in Computer Software Engineering — *Aug 2024 – May 2026*

Vellore Institute of Technology — Tamil Nadu, India
Bachelor of Technology (B.Tech) in Information Technology — *Jul 2017 – Jul 2021*